ATTENTION DEFICIT/HYPERACTIVITY DISORDER AND URINARY METABOLITES OF ORGANOPHOSPHATE PESTICIDES IN U.S. CHILDREN 8–15 YEARS

Context Exposure to organophosphate (OP) pesticides is common, and although these compounds have known neurotoxic properties, few studies examined risks for children in the general population. Objective To examine the association between the concentrations of urinary dialkyl phosphate (DAP) metabolites of OPs and attention deficit/hyperactivity disorder (ADHD) in children age 8 to 15 years. Participants and Methods Cross-sectional data from the National Health and Nutrition Examination Survey (2000–2004) were available for 1,139 children representative of the general U.S. population. A structured interview with a parent was used to ascertain ADHD diagnostic status, based on slightly modified criteria of the Diagnostic and Statistical Manual of Mental Disorders-IV. Results One hundred nineteen children met the diagnostic criteria for ADHD. Children with higher concentrations of urinary DAPs, especially dimethyl alkylphosphates (DMAP), were more likely to be diagnosed with ADHD. A 10-fold increase in DMAP concentration was associated with an odds ratio (OR) of 1.55 (95% confidence intervals [CI], 1.14–2.10), after adjusting for sex, age, race/ethnicity, poverty-income ratio, fasting duration, and urinary creatinine concentration. For the most commonly detected DMAP metabolite, dimethylthiophosphate, children with levels higher than the median of detectable concentrations had double the odds of ADHD (adjusted OR, 1.93 [95% CI, 1.23–3.02]) compared with those with non-detectable levels. Conclusions These findings support the hypothesis that OP exposure, at levels common in U.S. children, may contribute to ADHD prevalence. Prospective studies are needed to establish whether this association is causal.